1. Implicit and explicit analogues of common methods

(a)

We can write the Butcher tableau for Gauss–Legendre methods

$$\begin{array}{c|cc} c & A \\ & b^T \end{array} = \begin{array}{ccc} \frac{1}{2}-\frac{\sqrt{3}}{6} & \frac{1}{4} & \frac{1}{4}-\frac{\sqrt{3}}{6} \\ \frac{1}{2}+\frac{\sqrt{3}}{6} & \frac{1}{4}+\frac{\sqrt{3}}{6} & \frac{1}{4} \\ & \frac{1}{2} & \frac{1}{2} \end{array}$$

$$A = \begin{pmatrix} \frac{1}{4} & \frac{1}{4}-\frac{\sqrt{3}}{6} \\ \frac{1}{4}+\frac{\sqrt{3}}{6} & \frac{1}{4} \end{pmatrix},\ b = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}.$$

Assume the eigenvalue of f is λ, which means that we can assume

$$\frac{du}{dt} = f(u) = \lambda u$$

Plug this into implicit Runge-Kutta method and get

$$\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = f\left(\begin{pmatrix} u_n \\ u_n \end{pmatrix} + A\Delta t \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}\right) = \lambda \begin{pmatrix} u_n \\ u_n \end{pmatrix} + \lambda \Delta t A \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}$$

Which means that

$$\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = (I - \lambda \Delta t A)^{-1} \lambda \begin{pmatrix} u_n \\ u_n \end{pmatrix}$$

Take this into update formula and get

$$u_{n+1} = u_n + \Delta t b^T \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \left(1 + \lambda \Delta t b^T (I - \lambda \Delta t A)^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) u_n = \rho(z) u_n$$

Here $\rho(z) = 1 + z b^T (I - zA)^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

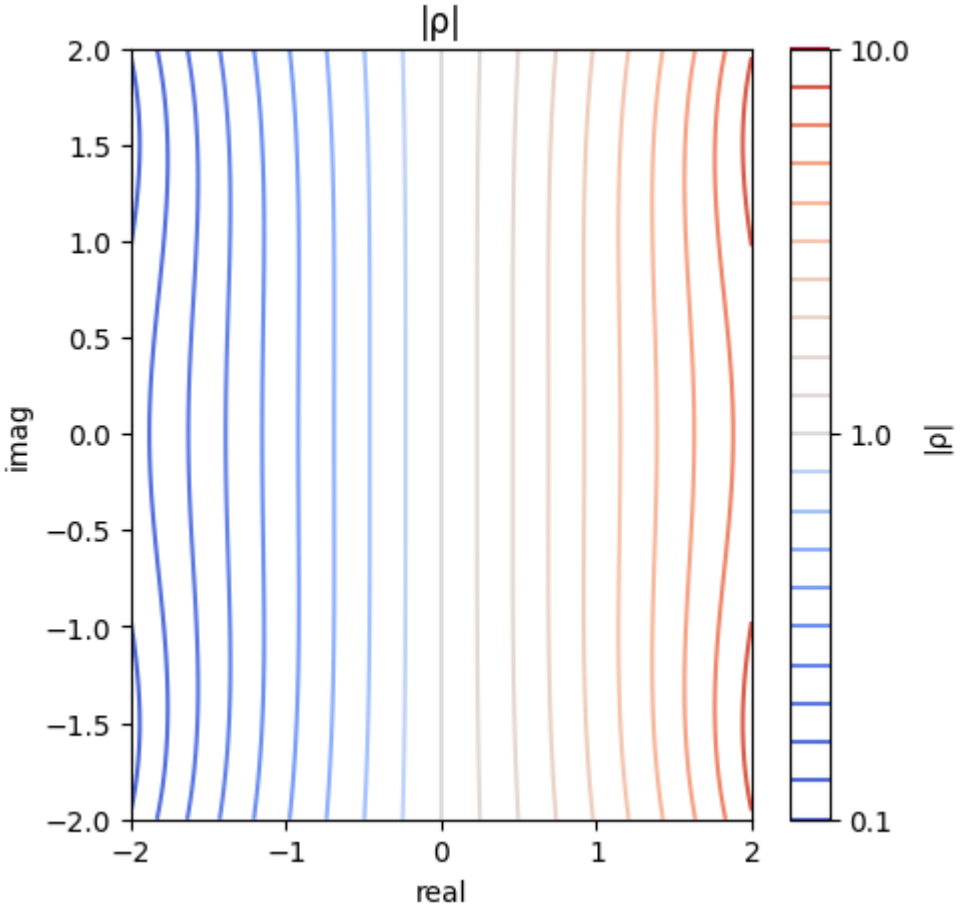
Stability requires $|\rho(z)| < 1$.

Let $z = \lambda \Delta t$, and $det = |I - zA| = \frac{1}{12} z^2 - \frac{1}{2} z + 1$.

Take A and b in and get

$$\rho(z) = 1 + \frac{z}{det} = \frac{\frac{1}{12} z^2 + \frac{1}{2} z + 1}{\frac{1}{12} z^2 - \frac{1}{2} z + 1}$$

We can plot the stability region on the complex plain



We can find that it is A-stable because $|\rho(z)| < 1$ when $\mathrm{Re}(z) < 0$, and it is not L-stable because $\lim_{z \to -\infty} \rho(z) = 1 \neq 0$.

I will not recommend using this fourth-order scheme in practice because it is fully implicit, which means I need to solve nonlinear system each step.

(b)

$$\frac{du}{dt} = f(u) = u_n{}'$$

Expand $f(u)$ at u_n to express u_{n+1}, u_{n-1}, u_{n-2}, etc

$$u_{n+1} = u_n + \Delta t u_n{}' + \frac{\Delta t^2}{2} u_n{}'' + \frac{\Delta t^3}{6} u_n{}''' + \cdots$$

$$u_{n-1} = u_n - \Delta t u_n{}' + \frac{\Delta t^2}{2} u_n{}'' - \frac{\Delta t^3}{6} u_n{}''' + \cdots$$

$$u_{n-2} = u_n - 2\Delta t u_n{}' + 2\Delta t^2 u_n{}'' - \frac{4\Delta t^3}{3} u_n{}''' + \cdots$$

Second-order method (s = 1)

$$u_{n+1} + \alpha_0 u_n + \alpha_1 u_{n-1} = \beta f(u_n)$$

$$\begin{aligned} \text{LHS} = \quad & u_{n+1} + \alpha_0 u_n + \alpha_1 u_{n-1} \\ = \quad & (1 + \alpha_0 + \alpha_1) u_n + (1 - \alpha_0) \Delta t u_n{}' + \left(\frac{1}{2} + \frac{\alpha_1}{2}\right) \Delta t^2 u_n{}'' + \left(\frac{1}{6} - \frac{\alpha_1}{6}\right) \Delta t^3 u_n{}''' + \cdots \end{aligned}$$

Match coefficients and get

$$\begin{cases} \alpha_0 = 0 \\ \alpha_1 = -1 \\ \beta = 2\Delta t \end{cases}$$

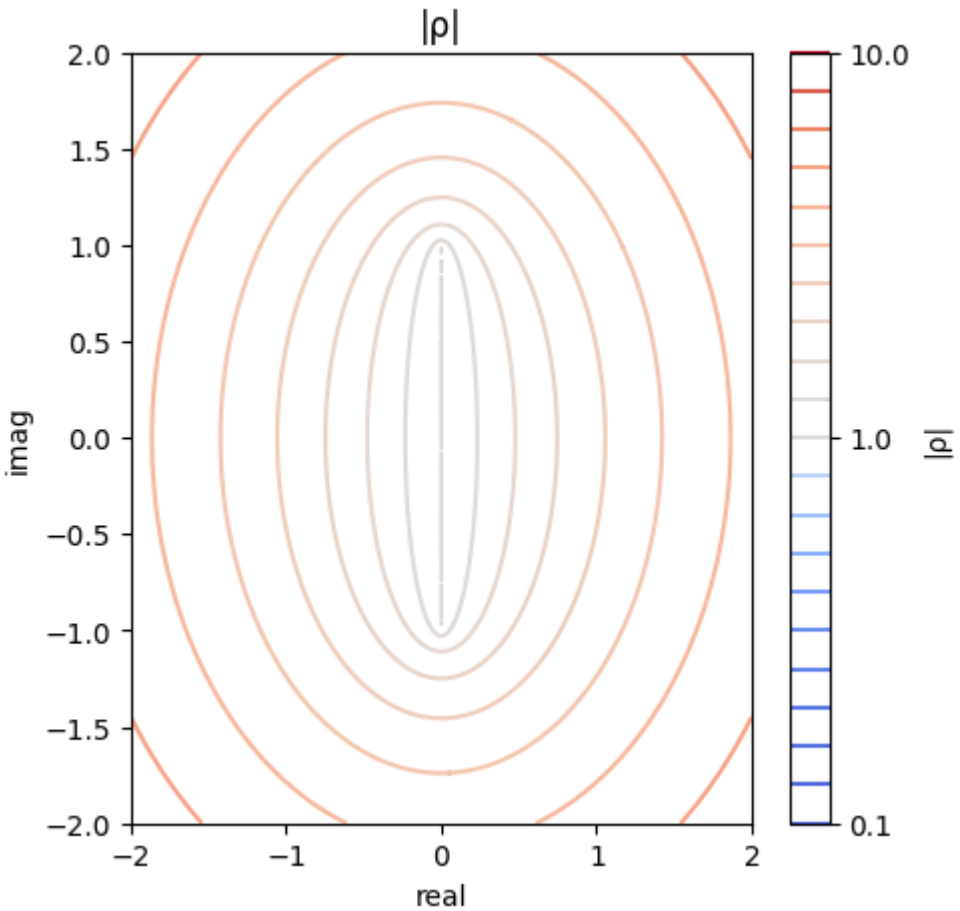
Second-order method should be

$$u_{n+1} - u_{n-1} = 2\Delta t f(u_n)$$

If we take $f(u_n) = \lambda u_n$, then the growth rate ρ can be obtained from

$$\rho^2 - 2z\rho - 1 = 0$$

Stability requires all the root $|\rho(z)| < 1$.



We can find that stability condition can only be satisfied when $|\mathrm{Imag}(z)| < 1$.

Third-order method (s = 2)

$$u_{n+1} + \alpha_0 u_n + \alpha_1 u_{n-1} + \alpha_2 u_{n-2} = \beta f(u_n)$$

$$\begin{aligned} \text{LHS} = \quad & u_{n+1} + \alpha_0 u_n + \alpha_1 u_{n-1} + \alpha_2 u_{n-2} \\ = \quad & (1 + \alpha_0 + \alpha_1 + \alpha_2) u_n + (1 - \alpha_0 - 2\alpha_2)\Delta t u_n' + \left(\frac{1}{2} + \frac{\alpha_1}{2} + 2\alpha_2\right)\Delta t^2 u_n'' + \left(\frac{1}{6} - \frac{\alpha_1}{6} - \frac{4}{3}\alpha_2\right)\Delta t^3 u_n''' + \cdots \end{aligned}$$

Match coefficients and get

$$\begin{cases} \alpha_0 = \dfrac{3}{2} \\ \alpha_1 = -3 \\ \alpha_2 = \dfrac{1}{2} \\ \beta = 3\Delta t \end{cases}$$

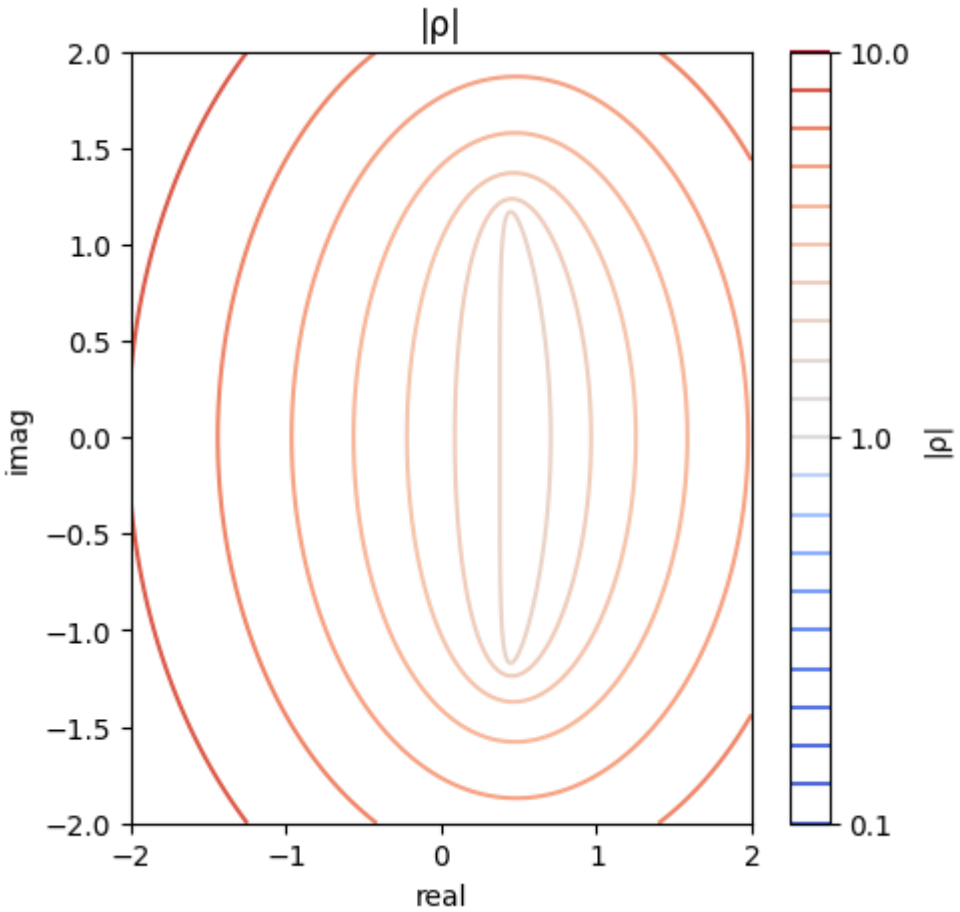
Third-order method should be

$$u_{n+1} + \frac{3}{2}u_n - 3u_{n-1} + \frac{1}{2}u_{n-2} = 3\Delta t f(u_n)$$

If we take $f(u_n) = \lambda u_n$, then the growth rate ρ can be obtained from

$$\rho^3 + \left(\frac{3}{2} - 3z\right)\rho^2 - 3\rho + \frac{1}{2} = 0$$

Stability requires all the root $|\rho(z)| < 1$.



We can find that none of the point on the complex can make this method stable. The min $\rho \approx 1.73$. I will not use these methods. They are not stable.